

8	(a) (i)	$p = h/\lambda$ $= (6.63 \times 10^{-34})/(6.50 \times 10^{-12})$ $= 1.02 \times 10^{-22} \text{ N s}$	C1 A1	[2]
	(ii)	$E = hc/\lambda \text{ or } E = pc$ $= (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(6.50 \times 10^{-12})$ $= 3.06 \times 10^{-14} \text{ J}$	C1 A1	[2]
	(b) (i)	$0.34 \times 10^{-12} = (6.63 \times 10^{-34})/(9.11 \times 10^{-31} \times 3.0 \times 10^8) \times (1 - \cos \theta)$ $\theta = 30.7^\circ$	C1 A1	[2]
	(ii)	deflected electron has energy this energy is derived from the incident photon deflected photon has less energy, longer wavelength (so $\Delta\lambda$ always positive)	M1 A1 B1	[3]